

GAZEMM: ROBUST OFF-ANGLE IRIS RECOGNITION VIA GAZE MANIFOLD MATCHING — SUPPLEMENTARY MATERIAL —

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This supplementary material provides the technical and experimental details that could not be included in the main paper because of the page limit. It is organized as follows:

- (A) Datasets
- (B) Implementation details
- (C) Comparison methods
- (D) Additional experimental results

Tables 1–5, Figs. 1–3, and Eqs. (1)–(8) refer to the main paper, whereas the tables and figures of this supplement are numbered with the prefix “S”.

A. DATASETS

Sec. 3.1 of the main paper lists the twelve datasets. Here we describe them in more detail.

- Fig. S1 shows a representative image from each dataset. The eight frontal datasets (CASIA v1.0, CASIA-Iris-Interval, CASIA-Iris-Thousand, CASIA-Iris-Lamp [1], PolyU Iris DB [2], MMU-v1 [3], and the Cogent and Vista subsets of the IIITD Contact Lens Iris DB (IIITD-CLI) [4]) are near-infrared datasets acquired under co-operative, frontal-gaze conditions. They differ in sensor, resolution, illumination, and pupil dilation, and thus cover a wide range of image quality.
- The four off-angle datasets were acquired under complementary conditions. EyeMovement [5] consists of eye-movement videos recorded while the subjects continuously tracked visual targets on a screen. OpenEDS [6] was captured at close range and off-axis by cameras embedded in a VR headset. CASIA-Iris-Degradation [1] contains controlled subsets for gaze, illumination, occlusion, and eye state. CASIA-Iris-Africa [7] covers large variations in gaze, illumination, and occlusion.
- Table S1 lists the number of images, the number of identities, and the average number of images per identity for each dataset. Identities are defined per eye, so the left and right eyes of the same person are treated as different

Table S1. Datasets used in the experiments. #IDs is the number of per-eye identities, and Img./ID is the average number of images per identity.

Dataset	#Images	#IDs	Img./ID
<i>Frontal</i>			
CASIA v1.0 (V1)	756	108	7.0
CASIA-Iris-Interval (Int)	2,639	393	6.7
CASIA-Iris-Thousand (Tho)	20,000	2,000	10.0
CASIA-Iris-Lamp (Lam)	16,212	819	19.8
PolyU Iris DB (Pol)	6,270	417	15.0
MMU-v1 (MMU)	450	88	5.1
IIITD-CLI Cogent (Cog)	1,163	202	5.8
IIITD-CLI Vista (Vis)	1,000	198	5.1
<i>Off-angle</i>			
EyeMovement (EM)	9,120	76	120.0
OpenEDS (OED)	27,431	186	147.5
CASIA-Iris-Degradation (Deg)	36,539	510	71.6
CASIA-Iris-Africa (Afr)	28,717	2,044	14.0

classes, and only comparisons between images of the same eye are counted as genuine.

B. IMPLEMENTATION DETAILS

The embedding network is an ImageNet-pretrained ResNet-34 with its classification layer replaced by a 512-dimensional embedding layer. Each grayscale periocular image is padded to a square using a constant gray value of 127, resized to 224×224 , replicated across three channels, and normalized to $[-1, 1]$. The resulting embedding is L_2 -normalized as in Eq. (2). Training uses batches of $P=16$ identities with $K=4$ images per identity (batch size 64) and the margin-free soft-plus batch-hard triplet loss in Eq. (3) [8]. The network is optimized with Adagrad (learning rate 0.075, weight decay 10^{-5}) for up to 150 epochs. Early stopping is based on validation EER with a patience of 10 epochs. For each identity, the gaze manifold is defined as the affine subspace centered at the mean embedding μ_i and spanned by the top $k=3$ eigenvectors of $A_i A_i^T$ in Eq. (4). We use fixed values of $k=3$ and $c=2.0$ across all twelve datasets without dataset-specific tuning.

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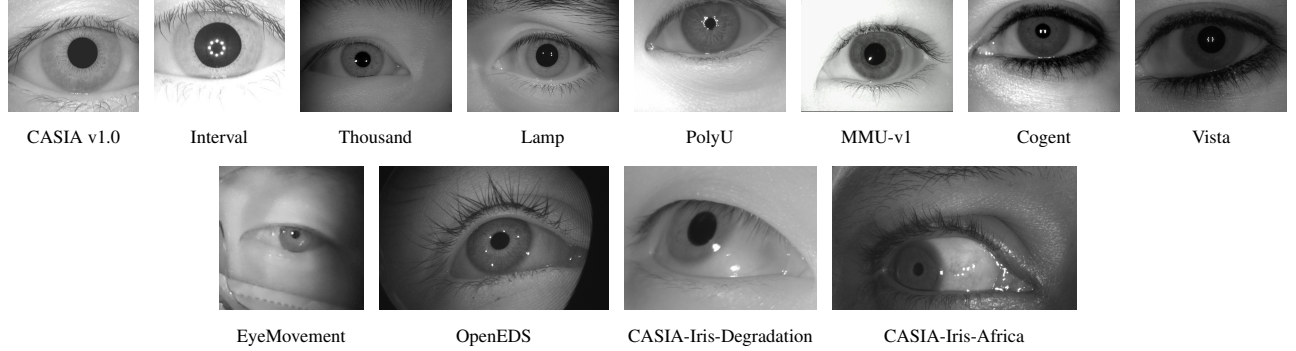


Fig. S1. Representative images from each dataset. Top: the eight frontal datasets. Bottom: the four off-angle datasets, which include geometric distortion caused by gaze deviation as well as variations in occlusion and illumination.

C. COMPARISON METHODS

All comparison methods are run with the authors’ released implementations and their default or recommended settings. As in the main paper, we report the equal error rate (EER), the false rejection rate (FRR) at a false acceptance rate of 10^{-3} , the Rank-1 accuracy, and the failure-to-enroll rate (FTE). Since the FTE of a method is determined by its segmentation or quality-check stage, we describe this stage for each method below.

- **WAHET** [9]: a real-time segmentation method that estimates the pupil and iris boundaries with a weighted adaptive Hough transform and an ellipsopolar transform. We run the official USITv3 (v3.0.0) chain, namely `wahet` (segmentation and rubber-sheet normalization to 512×64), `lg` (1-D log-Gabor code), and `hd` (minimum Hamming distance over ± 16 bit shifts). Every image is enrolled, so the FTE is zero.
- **USITv3** [10]: the open-source iris recognition toolkit of the University of Salzburg. We follow the combination recommended by the toolkit authors and used in [11], namely `caht` (contrast-adjusted Hough transform), `qsw` (quadratic spline wavelet code [12]), and `hd` with the `-sf` option, which excludes pairs whose masks do not overlap at any shift instead of assigning them a distance of zero. As with WAHET, every image is enrolled and the FTE is zero.
- **OSIRIS** [13]: the BioSecure reference implementation (v4.1) with Viterbi-based boundary estimation, Gabor phase encoding, and fractional Hamming distance. We use the default `process.ini` of the official distribution (pupil diameter 50–160, iris diameter 160–280, 512×64 normalization, and the official filter bank). Images for which the boundary estimation fails are not enrolled, which accounts for its FTE. The all-pairs matching is vectorized for speed and reproduces the official scores exactly.
- **HDBIF** [14]: binarized statistical image features (BSIF) with filters retrained for the iris domain, computed on iris images normalized with the segmentation network of the Notre Dame Computer Vision Research Lab (CVRL). We use the released implementation and filter weights with masked fractional Hamming distance. The official code contains no quality check, so none is applied.
- **WCI** [15]: the open-source pipeline (open-iris) of Worldcoin, which consists of CNN segmentation, classical iris-code encoding, and Hamming-distance matching. We run the complete pipeline, including its built-in validators (off-gaze, occlusion, pupil-to-iris ratio, and sharpness thresholds), with the released segmentation weights, and match iris codes with the official `HammingDistanceMatcher` (rotation shift of 15, separate half matching, and normalized distance). Images rejected by the validators are not enrolled, which is the source of its FTE.
- **DGR** [16]: a method that represents and matches iris features as dynamic graphs for robustness to occlusion. Its inputs are normalized by open-iris without the validators, as for UE-UGCL and Maxout-BA below. It is evaluated zero-shot with the released single-scale weights and the official graph matching. Because its score is a graph similarity rather than a distance between embedding vectors, it cannot be combined with the proposed matching and is therefore absent from Table 3 of the main paper.
- **UE-UGCL** and **Maxout-BA** [17]: an embedding method that jointly learns uncertainty factors, and the baseline maxout CNN from the same repository. Since no trained weights are released, both are retrained on the training folds of the same splits as the proposed method with the official settings. The inputs are normalized by open-iris and resized to 128×128 . UE-UGCL additionally applies the 3×3 median filter of its enhancement module. Training uses the ArcFace margin softmax ($s=64$, $m=0.5$) with the Kullback–Leibler (KL) term ($\lambda=0.01$) and the UGCL curriculum for UE-UGCL, and SGD (initial learn-

ing rate 0.001, momentum 0.9, weight decay 0.001) with the learning rate divided by 10 at epochs 30, 50, and 70, for 100 epochs with a batch size of 300. Matching uses the cosine distance between L_2 -normalized embeddings. For a fair protocol, we deviate from the original papers in two respects. No external-data pretraining is used, and the checkpoint is selected by the validation EER, as for the proposed method, instead of taking the last epoch. Their published numbers are therefore not comparable with Tables 1 and 2 of the main paper. DGR, UE-UGCL, and Maxout-BA share the same open-iris normalization stage (without the validators), which is why their FTE values coincide.

- **ArcIris** and **TripletIris** [11]: CNN embeddings trained on iris images normalized with the CVRL segmentation (as in HDBIF) with the ArcFace loss (ResNet-100) and with a triplet loss (ConvNeXt-tiny), respectively. Both are evaluated zero-shot with the released weights. ArcIris calls the official `checkQuality` function (minimum pupil and iris radii of 12 and 16 pixels), which is the source of its FTE, whereas TripletIris has no quality check in the official code.

D. ADDITIONAL EXPERIMENTAL RESULTS

Per-dataset results on the off-angle datasets. Table 1 of the main paper reports averages over the four off-angle datasets. Table S4 (see the end of this supplement) gives the per-dataset EER, FRR, Rank-1 accuracy, and FTE. GazeMM has the lowest EER, the lowest FRR, and the highest Rank-1 on every dataset.

Per-dataset results on the frontal datasets. Table 1 of the main paper reports averages over the eight frontal datasets. Table S5 (see the end of this supplement) gives the per-dataset EER, FRR, Rank-1 accuracy, and FTE. Unlike on the off-angle datasets, GazeMM is not the best method on every frontal dataset, but it remains competitive on all of them. Only WCI and GazeMM keep the EER below 1.1% on all eight datasets, and GazeMM ranks first or second in EER on four of them. Overall, the performance of GazeMM is comparable to that of the best-performing existing methods under frontal conditions and, unlike theirs, is retained under off-angle conditions.

Ablation study on all off-angle datasets. Table 4 of the main paper reports the ablation study on CASIA-Iris-Degradation, and Table S2 repeats it on all four off-angle datasets. The point-to-point variant keeps the score normalization and the quality filtering, and the variant without quality filtering is trained and evaluated on all images (FTE 0%). Removing any single component increases the EER and the FRR above those of GazeMM on CASIA-Iris-Degradation and CASIA-Iris-Africa. On EyeMovement and OpenEDS, the differences between the variants are within the fold-to-fold variation.

Table S2. Component-wise ablation on the four off-angle datasets (%).

Variant	EM		OED		Deg		Afr	
	EER	FRR	EER	FRR	EER	FRR	EER	FRR
Point-to-point matching	0.93	5.06	1.56	62.05	4.84	43.67	2.46	17.05
w/o Score normalization	0.11	0.12	0.66	32.63	0.83	7.08	0.32	2.14
w/o Quality filtering	0.01	0.01	0.62	34.76	0.41	2.39	0.21	1.42
GazeMM (Ours)	0.03	0.03	0.61	34.92	0.38	2.34	0.16	0.65

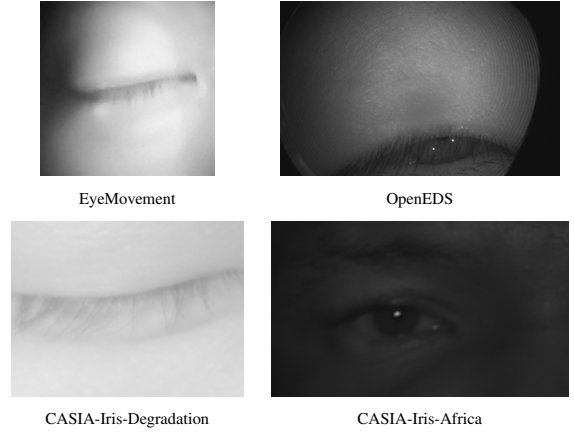


Fig. S2. Representative images rejected by quality filtering ($c=2.0$), one per off-angle dataset. The rejected images are blurred or captured with the eye closed, so that little or no iris texture is visible.

This shows that all three components contribute to the final performance. Manifold matching contributes by far the most, and score normalization and quality filtering add smaller gains.

Images rejected by quality filtering. Fig. S2 shows one representative image rejected by the quality filtering of Sec. 2.2.1 in the main paper for each of the four off-angle datasets. The rejected images are blurred or captured with the eye closed, so that little or no iris texture is visible. Because the threshold of Eq. (1) is a per-dataset statistic, the same rule rejects 0.0–6.4% of the images depending on the dataset. This fraction is reported as the FTE of GazeMM in Tables 1 and 2 of the main paper and in the FTE columns of Tables S4 and S5.

Effect of the quality-filtering parameter c . Table S3 gives the EER, FRR, and FTE on the four off-angle datasets for $c \in \{2.0, 3.0, 4.0\}$. The full experiment, including training, is repeated for each value. The filtering rejects 0.4–6.4% of the images at $c=2.0$, 0.0–3.1% at $c=3.0$, and 0.0–1.7% at $c=4.0$. The best value of c differs by dataset. EyeMovement performs best at $c=4.0$, CASIA-Iris-Degradation at $c=3.0$, and CASIA-Iris-Africa at $c=2.0$, while OpenEDS is insensitive to c . We nevertheless use a single value of $c=2.0$ for all twelve datasets. This value was fixed a priori by visual inspection of the rejected frames (Fig. S2) and was not tuned on the results,

Table S3. Effect of the quality-filtering parameter c on the four off-angle datasets (%). A larger c rejects fewer images. The main paper uses $c=2.0$ for all datasets.

Dataset	$c=2.0$			$c=3.0$			$c=4.0$		
	EER	FRR	FTE	EER	FRR	FTE	EER	FRR	FTE
EM	0.03	0.03	1.6	0.06	0.12	0.4	0.03	0.02	0.0
OED	0.61	34.92	0.4	0.62	34.76	0.0	0.62	34.76	0.0
Deg	0.38	2.34	3.1	0.37	2.01	0.2	0.41	2.39	0.0
Afr	0.16	0.65	6.4	0.20	1.79	3.1	0.19	1.42	1.7

so quality filtering serves as a safeguard against low-quality inputs rather than a tuned component.

E. REFERENCES

- [1] CASIA, “CASIA iris image databases (V1.0, V4, and Degradation V1.0),” [Online]. Available: <http://biometrics.idealtest.org>.
- [2] A. Kumar and A. Passi, “Comparison and combination of iris matchers for reliable personal authentication,” *Pattern Recognit.*, vol. 43, no. 3, pp. 1016–1026, 2010.
- [3] Multimedia University, “MMU iris image database,” 2004. [Online]. Available: <http://pesona.mmu.edu.my>.
- [4] N. Kohli, D. Yadav, M. Vatsa, and R. Singh, “Revisiting iris recognition with color cosmetic contact lenses,” in *Proc. Int. Conf. Biometrics (ICB)*, 2013, pp. 1–7.
- [5] National Information Society Agency (NIA), “Eye movement video data (AI-Hub),” Dataset no. 548, 2022. [Online]. Available: <https://aihub.or.kr>.
- [6] S. J. Garbin, Y. Shen, I. Schuetz, R. Cavin, G. Hughes, and S. S. Talathi, “OpenEDS: Open eye dataset,” arXiv:1905.03702, 2019.
- [7] J. Muhammad, Y. Wang, J. Hu, K. Zhang, and Z. Sun, “CASIA-Iris-Africa: A large-scale African iris image database,” *Mach. Intell. Res.*, vol. 21, no. 2, pp. 383–399, 2024.
- [8] A. Hermans, L. Beyer, and B. Leibe, “In defense of the triplet loss for person re-identification,” arXiv:1703.07737, 2017.
- [9] A. Uhl and P. Wild, “Weighted adaptive Hough and ellipsopolar transforms for real-time iris segmentation,” in *Proc. Int. Conf. Biometrics (ICB)*, 2012, pp. 283–290.
- [10] C. Rathgeb, A. Uhl, P. Wild, and H. Hofbauer, “Design decisions for an iris recognition SDK,” in *Handbook of Iris Recognition*, pp. 359–396. Springer, 2nd edition, 2016.
- [11] S. K. Khan, P. J. Flynn, and A. Czajka, “Lowering the barrier to IREX participation: Open-source algorithms, toolkit, and benchmarking for iris recognition,” arXiv:2605.20735, 2026.
- [12] L. Ma, T. Tan, Y. Wang, and D. Zhang, “Personal identification based on iris texture analysis,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 12, pp. 1519–1533, Dec. 2003.
- [13] N. Othman, B. Dorizzi, and S. Garcia-Salicetti, “OSIRIS: An open source iris recognition software,” *Pattern Recognit. Lett.*, vol. 82, pp. 124–131, 2016.

- [14] A. Czajka, D. Moreira, K. Bowyer, and P. Flynn, “Domain-specific human-inspired binarized statistical image features for iris recognition,” in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2019, pp. 959–967.
- [15] Worldcoin, “Open IRIS: Iris recognition inference system,” 2023. [Online]. Available: <https://github.com/worldcoin/open-iris>.
- [16] M. Ren, Y. Wang, Z. Sun, and T. Tan, “Dynamic graph representation for occlusion handling in biometrics,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2020, pp. 11940–11947.
- [17] J. Wei, H. Huang, Y. Wang, R. He, and Z. Sun, “Towards more discriminative and robust iris recognition by learning uncertain factors,” *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 865–879, 2022.

Table S4. Per-dataset results on the four off-angle datasets (%). Abbreviations are as in Table S1. The EER and FTE values are those of Table 2 in the main paper. **Bold** and underlined values denote the best and second-best EER, FRR, and Rank-1 per column.

Method	EM				OED				Deg				Afr			
	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE
WAHET [9]	35.98	96.36	18.26	0.0	38.32	99.68	1.18	0.0	29.23	99.62	0.30	0.0	39.80	85.46	22.66	0.0
USITv3 [10]	33.14	95.44	46.77	0.0	37.59	93.86	66.61	0.0	25.56	59.46	58.89	0.0	36.66	78.08	28.18	0.0
OSIRIS [13]	22.94	95.13	43.79	32.8	23.27	90.85	88.36	1.8	27.30	83.44	76.29	1.3	23.21	62.28	39.82	3.0
HDBIF [14]	27.35	71.75	84.52	0.0	28.49	89.37	87.96	0.0	26.61	54.00	87.32	0.0	22.57	57.97	48.42	0.0
WCI [15]	12.82	63.95	<u>99.15</u>	39.4	<u>6.27</u>	88.38	92.81	26.3	<u>5.06</u>	<u>11.29</u>	<u>98.04</u>	25.2	<u>4.25</u>	<u>19.21</u>	<u>58.31</u>	67.7
DGR [16]	14.10	62.08	99.08	39.2	15.40	88.90	<u>93.62</u>	25.5	17.26	75.64	96.34	13.9	23.79	63.76	46.81	3.3
UE-UGCL [17]	<u>10.70</u>	<u>57.26</u>	98.72	39.2	11.61	<u>87.44</u>	92.94	25.5	13.16	63.92	95.84	13.9	12.45	56.57	48.91	3.3
Maxout-BA [17]	15.10	69.42	97.79	39.2	27.62	90.87	90.96	25.5	18.64	77.78	93.76	13.9	27.16	86.49	34.65	3.3
Arclris [11]	15.64	67.13	98.22	2.9	16.20	88.21	92.57	0.1	16.54	37.91	92.75	1.7	17.69	44.46	52.37	0.1
TripletIris [11]	14.94	78.42	97.45	0.0	11.98	87.55	92.36	0.0	17.98	90.12	91.36	0.0	17.60	72.45	46.68	0.0
GazeMM (Ours)	0.03	0.03	100.00	1.6	0.61	34.92	99.50	0.4	0.38	2.34	99.62	3.1	0.16	0.65	58.48	6.4

Table S5. Per-dataset results on the eight frontal datasets (%). Abbreviations are as in Table S1. **Bold** and underlined values denote the best and second-best EER, FRR, and Rank-1 per column.

Method	V1				Int				Tho				Lam			
	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE
WAHET [9]	1.56	8.61	98.59	0.0	1.87	4.73	98.02	0.0	8.98	52.50	75.66	0.0	15.62	86.85	79.20	0.0
USITv3 [10]	<u>0.26</u>	<u>0.26</u>	100.00	0.0	0.67	0.77	99.68	0.0	22.47	46.27	81.20	0.0	18.43	38.38	61.12	0.0
OSIRIS [13]	0.60	0.97	100.00	0.0	0.89	1.16	99.86	0.0	6.60	21.35	96.79	0.2	4.66	13.61	98.92	0.1
HDBIF [14]	0.67	1.21	100.00	0.0	0.89	1.41	99.92	0.0	4.13	10.56	99.45	0.0	2.10	4.07	99.74	0.0
WCI [15]	0.00	0.00	100.00	1.7	0.00	0.00	100.00	1.0	<u>0.49</u>	<u>0.68</u>	99.91	7.7	<u>0.12</u>	<u>0.14</u>	<u>99.99</u>	2.2
DGR [16]	2.61	7.34	99.33	0.9	1.65	5.73	99.67	0.8	3.18	14.51	99.07	7.6	4.93	27.59	99.33	1.8
UE-UGCL [17]	15.93	68.40	83.14	0.9	11.13	56.41	87.92	0.8	7.69	40.95	93.86	7.6	6.06	33.75	98.56	1.8
Maxout-BA [17]	22.60	81.11	74.33	0.9	17.54	74.49	79.22	0.8	10.59	54.72	88.77	7.6	10.11	49.77	96.94	1.8
Arclris [11]	0.29	0.38	100.00	0.0	0.32	<u>0.43</u>	<u>99.97</u>	0.0	1.32	3.25	99.78	0.0	0.84	1.40	99.94	0.0
TripletIris [11]	0.33	1.23	100.00	0.0	0.46	1.61	<u>99.97</u>	0.0	2.43	12.17	99.21	0.0	0.68	2.03	99.92	0.0
GazeMM (Ours)	0.38	2.14	<u>99.91</u>	2.9	<u>0.24</u>	1.08	99.92	3.5	0.22	0.54	<u>99.89</u>	0.1	0.05	0.02	100.00	4.0

Method	Pol				MMU				Cog				Vis			
	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE	EER	FRR	Rank-1	FTE
WAHET [9]	24.86	45.10	85.52	0.0	4.40	23.93	96.22	0.0	10.61	20.01	94.22	0.0	6.39	22.43	93.97	0.0
USITv3 [10]	25.00	43.05	78.65	0.0	3.23	5.30	97.95	0.0	13.36	20.57	87.22	0.0	5.60	7.19	92.96	0.0
OSIRIS [13]	14.87	36.64	87.30	10.1	6.12	15.11	97.98	0.9	1.62	2.48	99.25	0.3	1.63	2.05	99.50	0.0
HDBIF [14]	1.62	3.22	99.67	0.0	1.34	1.89	99.56	0.0	<u>0.19</u>	<u>0.33</u>	100.00	0.0	0.34	0.42	99.90	0.0
WCI [15]	<u>0.79</u>	<u>0.97</u>	99.65	33.9	0.02	0.00	100.00	0.0	0.00	0.00	100.00	2.5	0.00	0.00	100.00	84.1
DGR [16]	5.02	20.08	99.29	0.7	1.55	5.30	99.56	0.0	2.51	8.43	98.85	0.0	4.05	9.10	98.80	0.0
UE-UGCL [17]	5.67	26.51	97.78	0.7	12.72	68.56	87.56	0.0	12.69	58.03	86.92	0.0	12.02	55.35	83.90	0.0
Maxout-BA [17]	9.05	41.45	96.30	0.7	16.49	76.15	82.96	0.0	20.88	75.64	75.03	0.0	19.96	75.13	68.03	0.0
Arclris [11]	0.99	1.67	99.63	0.0	1.08	1.74	99.56	0.0	0.00	0.00	100.00	0.0	0.35	0.45	99.90	0.0
TripletIris [11]	1.07	3.02	<u>99.70</u>	0.0	<u>0.58</u>	<u>1.48</u>	<u>99.93</u>	0.0	0.89	3.26	99.66	0.0	<u>0.20</u>	<u>0.37</u>	100.00	0.0
GazeMM (Ours)	0.18	0.33	99.79	3.5	1.04	6.47	99.38	4.0	0.60	4.21	<u>99.94</u>	0.0	0.28	0.83	<u>99.97</u>	3.3